Noir Atelier — Complete Database Schema Design

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Schema Architecture Overview

Noir Atelier's database is organized into **7 logical domains**, each supporting a core business capability. The schema prioritizes:

Design Principles

- **Normalization**: 3NF for transactional tables; denormalization where justified by query patterns
- **Auditability**: Every customer-facing decision (refund, exchange, escalation) is logged immutably
- **Scalability**: Partitioning by date for time-series tables; sharding-ready keys (customer_id) in all tables
- **Data Integrity**: Foreign keys enforced; constraints prevent invalid state transitions
- **Performance**: Strategic indexes on query paths; JSON columns for flexible attributes

Database Domains

Domain	Purpose	Key Tables	Agents Access
Catalog	Products, inventory, categories	products, product_variants, inventory, categories	Catalog Agent (read)
Customers & Auth	User accounts, profiles, authentication	customers, addresses, payment_methods, users, roles	Order Agent, Support Agent
Orders & Transactions	Order lifecycle, payment, invoicing	orders, order_items, payments, invoices	Order Agent, Payment Agent
Fulfillment	Shipping, tracking, logistics	shipments, shipment_tracking, courier_partners	Fulfillment Agent

Domain	Purpose	Key Tables	Agents Access
Returns & Remedies	Exchanges, refunds, damage assessment	returns, return_items, refunds, damage_assessments	Returns Agent, Compliance Agent
Support & Communication	Customer inquiries, tickets, messages	support_tickets, ticket_messages, knowledge_base	Support Agent
Configuration & Compliance	Policies, tax rules, business config	policies, tax_rules, coupon_codes, gstin_config	Compliance Agent, Admin Agent
Audit & Observability	Agent decisions, compliance logs, errors	audit_log, agent_decisions, compliance_flags, system_errors	All agents (write)

Key Relationships

```
Customers (root)
  ├─ Addresses (many)
  ├─ Payment Methods (many)
  ├─ Orders (many)
  |  ├─ Order Items (many)
  |  ├─ Payments (many)
  |  ├─ Invoices (many)
  |  ├─ Shipments (many)
  |  |  ├─ Shipment Tracking Events (many)
  |  |  └─ Courier Partner
  |  └─ Returns (many)
  |     ├─ Return Items (many)
  |     └─ Refunds (many)
  └─ Support Tickets (many)
     └─ Ticket Messages (many)

Products (root)
  ├─ Categories (many)
  ├─ Product Variants (many)
  |  └─ Inventory (one-to-one)
  └─ Reviews (many)
```

Data Types & Storage Strategy

- **VARCHAR(255)** for names, emails, short codes (SKU, order_id)
- **TEXT** for descriptions, policy text, long content
- **JSON** for flexible attributes (product options, policy data, error details)
- **DECIMAL(10,2)** for monetary values (₹)
- **TIMESTAMP WITH TIME ZONE** for all dates (audit trail, order timestamps)
- **UUID** for public IDs (customer_id, order_id) while auto-incrementing integers for internal use
- **ENUM** for fixed states (order_status, return_status, payment_method_type)
- **BYTEA** for encrypted data (sensitive customer fields)

Soft Deletes vs Hard Deletes

- **Soft deletes** (deleted_at timestamp): customers, products, coupons (audit trail, referential integrity)
- **Hard deletes** never used (compliance: retain data for 7 years for tax/dispute resolution)
- **Cascading deletes** only for purely derived data (shipment_tracking events when shipment deleted)

Temporal Aspects

- **created_at, updated_at** on every table (UTC timestamps)
- **effective_date, expires_at** for policy versions, coupon codes
- **event_timestamp** for all audit/tracking events (millisecond precision)
- **Partitioning**: orders, audit_log, shipment_tracking partitioned by month

Constraints & Validation

- **Primary Keys**: UUID or auto-increment integer (never business keys)
- **Unique Constraints**: email (customers), order_number, sku (products)
- **Check Constraints**: price > 0, quantity >= 0, status IN (...)
- **Not Null**: Essential fields (customer_id, order_id, created_at, amount)
- **Foreign Keys**: Enforced with ON DELETE RESTRICT (prevent orphans) or ON DELETE CASCADE (for audit)

Multi-tenancy & Data Isolation

- Single-tenant for Phase 1-4 (customer_id in all transactional tables for sharding readiness)
- Business config (tax_rules, policies) global but versioned

- Audit logs queryable per customer for support/compliance

Complete Table Reference (32 Tables)

1. CUSTOMERS (Root entity)

Purpose: User accounts, authentication, profile management

Column	Type	Constraints	Notes
customer_id	UUID	PK	Immutable, public-facing order reference
email	VARCHAR(255)	UNIQUE, NOT NULL	Lowercase, indexed for search
phone	VARCHAR(15)	UNIQUE	Indian format: 10 digits
password_hash	BYTEA	NOT NULL	bcrypt, never stored plain
first_name	VARCHAR(100)	NOT NULL	
last_name	VARCHAR(100)		
date_of_birth	DATE		For age verification if needed
gender	ENUM('M','F','Other','Prefer not to say')		
email_verified	BOOLEAN	DEFAULT FALSE	Customer must verify email
phone_verified	BOOLEAN	DEFAULT FALSE	Verify via OTP
newsletter_opt_in	BOOLEAN	DEFAULT FALSE	DPDP Act: explicit consent
marketing_consent	BOOLEAN	DEFAULT FALSE	Separate consent for marketing
preferred_language	VARCHAR(5)	DEFAULT 'en'	en, hi, ta, te
loyalty_points	INTEGER	DEFAULT 0	Redeemable points
vip_status	ENUM('standard','silver','gold','platinum')	DEFAULT 'standard'	Based on lifetime value

Column	Type	Constraints	Notes
last_login_at	TIMESTAMP		For account security audit
created_at	TIMESTAMP	NOT NULL	UTC
updated_at	TIMESTAMP	NOT NULL	UTC
deleted_at	TIMESTAMP		Soft delete for GDPR

2. ADDRESSES

Purpose: Billing, shipping, return addresses per customer

Column	Type	Constraints	Notes
address_id	BIGINT	PK	
customer_id	UUID	FK -> customers	NOT NULL, indexed
address_type	ENUM('billing','shipping','return')	NOT NULL	Customer can have multiple
full_name	VARCHAR(100)	NOT NULL	Recipient name
street_address	VARCHAR(255)	NOT NULL	
apartment_floor	VARCHAR(100)		Apt no, floor, building name
locality	VARCHAR(100)	NOT NULL	Neighborhood
city	VARCHAR(100)	NOT NULL	
state	VARCHAR(100)	NOT NULL	Tamil Nadu, etc.
postal_code	VARCHAR(10)	NOT NULL	PIN code, 6 digits
country	VARCHAR(2)	DEFAULT 'IN'	ISO 3166-1 alpha-2
phone	VARCHAR(15)	NOT NULL	Delivery contact
landmark	VARCHAR(100)		"Near mosque", "behind mall"

Column	Type	Constraints	Notes
google_maps_url	TEXT		For GPS coordinates
latitude	DECIMAL(10,8)		For precise delivery
longitude	DECIMAL(11,8)		
is_default_shipping	BOOLEAN	DEFAULT FALSE	Pre-select in checkout
is_serviceable	BOOLEAN	DEFAULT TRUE	Agent verifies against PIN list
created_at	TIMESTAMP	NOT NULL	
updated_at	TIMESTAMP	NOT NULL	

3. PAYMENT_METHODS

Purpose: Stored payment credentials (tokenized, not raw card data)

Column	Type	Constraints	Notes
payment_method_id	UUID	PK	
customer_id	UUID	FK -> customers	NOT NULL
method_type	ENUM('upi','card','netbanking','bank_transfer')	NOT NULL	
nickname	VARCHAR(50)		"My HDFC Debit Card"
is_default	BOOLEAN	DEFAULT FALSE	Auto-select in checkout
token	VARCHAR(255)	NOT NULL	Razorpay/PayU token, encrypted
last_four	VARCHAR(4)		For display: "****1234"
card_network	VARCHAR(20)		Visa, Mastercard, RuPay
expiry_date	VARCHAR(5)		MM/YY

Column	Type	Constraints	Notes
bank_name	VARCHAR(100)		HDFC, ICICI, etc. (for netbanking)
account_number	VARCHAR(255)	ENCRYPTED	Never stored plain
ifsc_code	VARCHAR(11)		For bank transfers
verified	BOOLEAN	DEFAULT TRUE	Payment gateway verified
created_at	TIMESTAMP	NOT NULL	
deleted_at	TIMESTAMP		Soft delete

4. PRODUCTS

Purpose: Product catalog, metadata

Column	Type	Constraints	Notes
product_id	UUID	PK	
sku	VARCHAR(50)	UNIQUE, NOT NULL	NA-MEN-001 format
name	VARCHAR(255)	NOT NULL	"Noir Essential Oversized Tee"
slug	VARCHAR(255)	UNIQUE	URL-friendly: noir-essential-tee
description	TEXT		Full product description
short_description	VARCHAR(500)		For listings
category_id	BIGINT	FK -> categories	NOT NULL
subcategory_id	BIGINT	FK -> categories	Optional, for nested categories
price_inr	DECIMAL(10,2)	NOT NULL, CHECK > 0	Base price

Column	Type	Constraints	Notes
compare_at_price_inr	DECIMAL(10,2)		Crossed-out "was" price
cost_price_inr	DECIMAL(10,2)		Internal cost (not shown)
gst_rate	DECIMAL(4,2)	DEFAULT 5.0	0, 5, 12, 18% for different items
gst_hsn_code	VARCHAR(8)		6204 for shirts, 6203 for pants
material	VARCHAR(100)		Cotton, Linen, Polyester
material_percentage	JSON		{"cotton": 100}, {"cotton": 60, "linen": 40}
care_instructions	TEXT		"Wash at 30°C", machine wash, etc.
brand	VARCHAR(100)		In-house brand = "Noir Atelier"
designer_name	VARCHAR(100)		If applicable
made_in_country	VARCHAR(2)		ISO code, IN for India
made_in_city	VARCHAR(100)		Tiruppur, Bangalore, etc.
manufacturing_date	DATE		Batch tracking
weight_grams	INTEGER		For shipping calculation
dimensions_length_cm	DECIMAL(6,2)		For packaging
dimensions_width_cm	DECIMAL(6,2)		
dimensions_height_cm	DECIMAL(6,2)		
is_active	BOOLEAN	DEFAULT TRUE	Soft control over visibility
is_featured	BOOLEAN	DEFAULT FALSE	Show on homepage
bestseller_rank	INTEGER		NULL if not bestseller, 1=top
rating	DECIMAL(3,2)	DEFAULT 0	Aggregated from reviews
review_count	INTEGER	DEFAULT 0	

Column	Type	Constraints	Notes
external_product_url	TEXT		Link to supplier/manufacturer
created_at	TIMESTAMP	NOT NULL	
updated_at	TIMESTAMP	NOT NULL	
deleted_at	TIMESTAMP		Soft delete

5. PRODUCT_VARIANTS

Purpose: Colors, sizes, SKU per variant

Column	Type	Constraints	Notes
variant_id	BIGINT	PK	
product_id	UUID	FK -> products	NOT NULL
variant_sku	VARCHAR(50)	UNIQUE, NOT NULL	NA-MEN-001-BLK-M (product-color-size)
color_name	VARCHAR(50)	NOT NULL	Black, Navy, Maroon
color_hex	VARCHAR(7)		#000000 for display
size	VARCHAR(10)	NOT NULL	S, M, L, XL, XXL, or numeric (2Y, 3Y)
size_chart_url	TEXT		Link to size guide
price_inr	DECIMAL(10,2)		Override product base price if different
quantity_on_hand	INTEGER	DEFAULT 0	De-normalized from inventory (denormalized for performance; authoritative source is inventory table)
variant_image_url	TEXT		Override product image for this color
is_available	BOOLEAN	DEFAULT TRUE	Display availability

Column	Type	Constraints	Notes
created_at	TIMESTAMP	NOT NULL	
updated_at	TIMESTAMP	NOT NULL	

6. INVENTORY

Purpose: Stock tracking per variant, reorder logic

Column	Type	Constraints	Notes
inventory_id	BIGINT	PK	
variant_id	BIGINT	FK -> product_variants	NOT NULL, UNIQUE (one-to-one)
quantity_on_hand	INTEGER	NOT NULL, DEFAULT 0	Actual stock
reserved_quantity	INTEGER	DEFAULT 0	In pending orders
available_quantity	INTEGER	GENERATED AS (quantity_on_hand - reserved_quantity)	Virtual column
reorder_level	INTEGER	DEFAULT 10	Reorder when stock drops below
reorder_quantity	INTEGER	DEFAULT 50	Order this many units
supplier_id	BIGINT	FK -> suppliers	
lead_time_days	INTEGER		How long reorder takes
last_reordered_at	TIMESTAMP		When last restocked
next_reorder_date	DATE		Calculated: last_reordered_at + lead_time_days
created_at	TIMESTAMP	NOT NULL	
updated_at	TIMESTAMP	NOT NULL	

7. CATEGORIES

Purpose: Product categories, hierarchy

Column	Type	Constraints	Notes
category_id	BIGINT	PK	
name	VARCHAR(100)	UNIQUE, NOT NULL	Men, Women, Kids
slug	VARCHAR(100)	UNIQUE	men, women, kids
description	TEXT		For category page
parent_category_id	BIGINT	FK -> categories	NULL for root (Men), subcategories have parent
image_url	TEXT		Category icon/banner
display_order	INTEGER		Sort order on homepage
is_active	BOOLEAN	DEFAULT TRUE	
created_at	TIMESTAMP	NOT NULL	
updated_at	TIMESTAMP	NOT NULL	

8. ORDERS

Purpose: Order header, lifecycle management

Column	Type	Constraints	Notes
order_id	UUID	PK	Public-facing order number
order_number	VARCHAR(20)	UNIQUE, NOT NULL	"NA-20261029-00142" for display
customer_id	UUID	FK -> customers	NOT NULL
order_status	ENUM('pending_payment','confirmed','processing','shipped','delivered','returned','cancelled')	DEFAULT 'pending_payment'	NOT NULL
subtotal_inr	DECIMAL(10,2)	NOT NULL	Sum of item prices
tax_inr	DECIMAL(10,2)	DEFAULT 0	GST total
shipping_charge_inr	DECIMAL(10,2)	DEFAULT 0	0 or 99

Column	Type	Constraints	Notes
discount_amount_inr	DECIMAL(10,2)	DEFAULT 0	Coupon/promo
total_inr	DECIMAL(10,2)	NOT NULL	subtotal + tax + shipping - discount
currency	VARCHAR(3)	DEFAULT 'INR'	ISO 4217
shipping_address_id	BIGINT	FK -> addresses	NOT NULL
billing_address_id	BIGINT	FK -> addresses	Usually same as shipping
delivery_date_estimate	DATE		Calculated at order time
actual_delivery_date	DATE		Filled when delivered
notes	TEXT		Customer special requests
agent_notes	TEXT		Internal notes (e.g., "Call before delivery")
created_at	TIMESTAMP	NOT NULL	UTC
updated_at	TIMESTAMP	NOT NULL	
confirmed_at	TIMESTAMP		When payment confirmed
shipped_at	TIMESTAMP		When dispatched
delivered_at	TIMESTAMP		When delivery confirmed
cancelled_at	TIMESTAMP		If cancelled
deleted_at	TIMESTAMP		Soft delete (rare)

9. ORDER_ITEMS

Purpose: Line items per order

Column	Type	Constraints	Notes
order_item_id	BIGINT	PK	
order_id	UUID	FK -> orders	NOT NULL
variant_id	BIGINT	FK -> product_variants	NOT NULL
product_name	VARCHAR(255)	NOT NULL	De-normalized for invoice
sku	VARCHAR(50)	NOT NULL	De-normalized
color	VARCHAR(50)		De-normalized

Column	Type	Constraints	Notes
size	VARCHAR(10)		De-normalized
quantity	INTEGER	NOT NULL, CHECK > 0	
unit_price_inr	DECIMAL(10,2)	NOT NULL	Price at time of order
tax_inr	DECIMAL(10,2)	DEFAULT 0	Line-item tax
discount_inr	DECIMAL(10,2)	DEFAULT 0	Specific to this item
total_inr	DECIMAL(10,2)	NOT NULL	(unit_price * quantity) + tax - discount
return_status	ENUM('none','requested','approved','received','completed','rejected')	DEFAULT 'none'	Track return state per item
created_at	TIMESTAMP	NOT NULL	

10. PAYMENTS

Purpose: Payment transaction records

Column	Type	Constraints	Notes
payment_id	UUID	PK	
order_id	UUID	FK -> orders	NOT NULL
payment_method_type	ENUM('upi','card','netbanking','bank_transfer','in_store')	NOT NULL	
payment_method_id	UUID	FK -> payment_methods	Nullable for one-time methods
amount_inr	DECIMAL(10,2)	NOT NULL, CHECK > 0	
payment_status	ENUM('pending','processing','confirmed','failed','refunded','partially_refunded')	DEFAULT 'pending'	NOT NULL
gateway_reference	VARCHAR(255)		Razorpay payment_id, external reference
gateway_response	JSONB		Full response from payment gateway (encrypted)
bank_transaction_id	VARCHAR(255)		For bank transfer verification

Column	Type	Constraints	Notes
upi_transaction_id	VARCHAR(255)		For UPI reconciliation
verification_status	ENUM('verified','pending_verification','failed_verification')	DEFAULT 'pending_verification'	Manual or auto-verified
verified_by	VARCHAR(255)		User/agent who verified
verified_at	TIMESTAMP		
error_message	TEXT		Why payment failed
retry_count	INTEGER	DEFAULT 0	Number of retry attempts
next_retry_at	TIMESTAMP		Schedule auto-retry
created_at	TIMESTAMP	NOT NULL	
completed_at	TIMESTAMP		When payment confirmed

11. INVOICES

Purpose: GST invoices, tax documents

Column	Type	Constraints	Notes
invoice_id	BIGINT	PK	
order_id	UUID	FK -> orders	UNIQUE (one invoice per order)
invoice_number	VARCHAR(20)	UNIQUE	Auto-generated per GST rules
invoice_date	DATE	NOT NULL	Dispatch date
gst_invoice_type	ENUM('regular','simplified','exempted')		Based on business status
seller_name	VARCHAR(255)	NOT NULL	"Noir Atelier"
seller_gstin	VARCHAR(15)		Updated from compliance config
seller_address	TEXT	NOT NULL	Complete address for GST
billing_customer_name	VARCHAR(255)	NOT NULL	
billing_customer_gstin	VARCHAR(15)		If B2B
billing_customer_address	TEXT	NOT NULL	

Column	Type	Constraints	Notes
shipping_address	TEXT		May differ from billing
line_items_json	JSONB		Array of items with HSN, tax rate, amount
subtotal_inr	DECIMAL(10,2)	NOT NULL	
sgst_inr	DECIMAL(10,2)	DEFAULT 0	SGST (same state)
cgst_inr	DECIMAL(10,2)	DEFAULT 0	CGST (central)
igst_inr	DECIMAL(10,2)	DEFAULT 0	IGST (inter-state)
total_tax_inr	DECIMAL(10,2)	NOT NULL	SGST + CGST + IGST
total_inr	DECIMAL(10,2)	NOT NULL	
qr_code	TEXT		E-invoice QR code (if required)
irn	VARCHAR(64)		Invoice Reference Number (IRN)
ack_num	VARCHAR(64)		Acknowledgement number from GST portal
ack_date_time	TIMESTAMP		When GST portal acknowledged
is_generated	BOOLEAN	DEFAULT FALSE	Auto-generate on dispatch
generated_at	TIMESTAMP		
pdf_url	TEXT		S3 URL to PDF
created_at	TIMESTAMP	NOT NULL	

12. SHIPMENTS

Purpose: Shipping arrangements, courier integration

Column	Type	Constraints	Notes
shipment_id	UUID	PK	

Column	Type	Constraints	Notes
order_id	UUID	FK -> orders	NOT NULL (one order may have multiple shipments if split)
courier_partner_id	BIGINT	FK -> courier_partners	NOT NULL
awb_number	VARCHAR(50)	UNIQUE	Airway Bill number from courier
tracking_url	TEXT		Link to courier tracking page
shipment_status	ENUM('created','picked_up','in_transit','out_for_delivery','delivered','failed','returned')	DEFAULT 'created'	
weight_grams	INTEGER		Actual weight of shipment
dimensions_json	JSONB		{"length_cm": 20, "width": 15, "height": 10}
shipment_charge_inr	DECIMAL(10,2)		What was paid to courier
shipping_address_id	BIGINT	FK -> addresses	NOT NULL
scheduled_pickup_date	DATE		When courier will pick up
actual_pickup_date	DATE		
scheduled_delivery_date	DATE		Estimated delivery
actual_delivery_date	DATE		When customer received
signature_proof_url	TEXT		Image/proof of delivery
delivery_notes	TEXT		"Left with security", etc.
failed_delivery_reason	TEXT		If delivery failed
return_to_warehouse	BOOLEAN	DEFAULT FALSE	RTO (Return to Origin)
created_at	TIMESTAMP	NOT NULL	
updated_at	TIMESTAMP	NOT NULL	

13. SHIPMENT_TRACKING

Purpose: Tracking events, GPS trail

Column	Type	Constraints	Notes
tracking_id	BIGINT	PK	
shipment_id	UUID	FK -> shipments	NOT NULL

Column	Type	Constraints	Notes
event_type	ENUM('created','picked_up','in_transit','out_for_delivery','delivered','failed','returned','lost','damaged')	NOT NULL	
event_description	TEXT	NOT NULL	"Picked up from warehouse", "Out for delivery", etc.
event_location	VARCHAR(255)		City or coordinates
latitude	DECIMAL(10,8)		GPS location
longitude	DECIMAL(11,8)		
event_timestamp	TIMESTAMP	NOT NULL	When event occurred (per courier)
received_at	TIMESTAMP	NOT NULL	When we received update
courier_event_id	VARCHAR(255)		External event ID
status_image_url	TEXT		Photo proof (if delivery photo)

14. COURIER_PARTNERS

Purpose: Integration with shipping providers

Column	Type	Constraints	Notes
courier_id	BIGINT	PK	
name	VARCHAR(100)	UNIQUE, NOT NULL	Shiprocket, Easypost, etc.
api_key	VARCHAR(255)	ENCRYPTED	Credentials for API calls
api_secret	VARCHAR(255)	ENCRYPTED	
api_base_url	TEXT		https://api.shiprocket.in
is_active	BOOLEAN	DEFAULT TRUE	Enable/disable courier

Column	Type	Constraints	Notes
priority_rank	INTEGER		1 = first choice for routing
serviceable_pincodes_json	JSONB		Cached list of serviceable PIN codes
rate_per_kg_inr	DECIMAL(10,2)		Pricing per kg
cod_fee_percent	DECIMAL(5,2)		Cash on delivery fee percentage
average_delivery_days	INTEGER		Expected delivery time
last_sync_at	TIMESTAMP		When we last updated their rates/coverage
created_at	TIMESTAMP	NOT NULL	

15. RETURNS

Purpose: Return/exchange requests

Column	Type	Constraints	Notes
return_id	UUID	PK	
order_id	UUID	FK -> orders	NOT NULL
return_request_type	ENUM('exchange','refund','repair','inspection')	NOT NULL	
return_reason	ENUM('defective','incorrect_item','size_mismatch','color_mismatch','quality_issue','damaged_in_shipping','changed_mind','other')	NOT NULL	
return_reason_details	TEXT		Customer explanation
return_status	ENUM('requested','approved','rejected','received','inspection_pending','completed','cancelled')	DEFAULT 'requested'	
return_request_date	DATE	NOT NULL	When customer initiated
eligibility_check	ENUM('eligible','outside_window','failed_condition','legal_remedy_applies')		Compliance check result
days_since_delivery	INTEGER		Auto-calculated for eligibility
pickup_date	DATE		When courier picks up
return_received_date	DATE		When item arrives at warehouse

Column	Type	Constraints	Notes
inspection_date	DATE		When we inspect the item
inspection_result	ENUM('approved_condition','damaged','counterfeit','tampered','missing_parts')		
approved_remedy	ENUM('exchange','refund','store_credit','repair')		Decision (may override customer request)
replacement_variant_id	BIGINT	FK -> product_variants	For exchange requests
replacement_order_id	UUID	FK -> orders	Replacement order created
refund_approved_amount_inr	DECIMAL(10,2)		May not be full amount (e.g., damage deduction)
refund_reason_code	VARCHAR(50)		For GST/tax adjustments
deduction_amount_inr	DECIMAL(10,2)	DEFAULT 0	Deduction for damage, wear, etc.
notes	TEXT		Return notes, condition description
photo_urls	TEXT[]		Photos of returned item
created_at	TIMESTAMP	NOT NULL	
completed_at	TIMESTAMP		

16. RETURN_ITEMS

Purpose: Line-by-line return tracking

Column	Type	Constraints	Notes
return_item_id	BIGINT	PK	
return_id	UUID	FK -> returns	NOT NULL
order_item_id	BIGINT	FK -> order_items	NOT NULL
quantity_returned	INTEGER	NOT NULL	How many of this item returned
unit_refund_amount_inr	DECIMAL(10,2)		Amount per item
total_refund_amount_inr	DECIMAL(10,2)		quantity_returned * unit_refund_amount
item_condition	ENUM('unused','worn','damaged','defective','counterfeit')		Assessment

Column	Type	Constraints	Notes
item_notes	TEXT		Description of condition

17. REFUNDS

Purpose: Refund transactions, traceability

Column	Type	Constraints	Notes
refund_id	UUID	PK	
return_id	UUID	FK -> returns	Nullable (some refunds not from returns)
order_id	UUID	FK -> orders	NOT NULL
refund_amount_inr	DECIMAL(10,2)	NOT NULL	
refund_reason	ENUM('customer_return','order_cancelled','payment_failed','duplicate_charge','dispute_resolution','goodwill')	NOT NULL	
refund_status	ENUM('initiated','processing','completed','failed','cancelled')	DEFAULT 'initiated'	
original_payment_method_id	UUID	FK -> payment_methods	Refund to same method
refund_gateway_reference	VARCHAR(255)		Gateway refund ID
refund_bank_transaction_id	VARCHAR(255)		
refund_initiated_at	TIMESTAMP	NOT NULL	
refund_completed_at	TIMESTAMP		When funds reached customer
processing_fee_inr	DECIMAL(10,2)	DEFAULT 0	Gateway fee (absorbed or deducted)
notes	TEXT		Reason details
approved_by	VARCHAR(255)		User/agent who approved
created_at	TIMESTAMP	NOT NULL	

18-22. Support System

18. SUPPORT_TICKETS

Column	Type	Constraints
ticket_id	UUID	PK
customer_id	UUID	FK -> customers
order_id	UUID	FK -> orders (nullable, may not be order-related)
subject	VARCHAR(255)	NOT NULL
description	TEXT	NOT NULL
ticket_status	ENUM('open','in_progress','waiting_customer','resolved','closed','reopened')	DEFAULT 'open'
priority	ENUM('low','medium','high','urgent')	DEFAULT 'medium'
category	ENUM('product_question','order_issue','shipping','return_exchange','payment','account','other')	NOT NULL
assigned_to_user_id	BIGINT	FK -> users (nullable)
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL
resolved_at	TIMESTAMP	
resolution_notes	TEXT	

19. TICKET_MESSAGES

Column	Type	Constraints
message_id	BIGINT	PK
ticket_id	UUID	FK -> support_tickets
sender_type	ENUM('customer','agent','system')	NOT NULL

Column	Type	Constraints
sender_user_id	UUID | BIGINT	Nullable (system generated messages)
message_text	TEXT	NOT NULL
attachments_json	JSONB	URLs to uploaded files
is_public	BOOLEAN	DEFAULT TRUE
created_at	TIMESTAMP	NOT NULL

20. KNOWLEDGE_BASE

Column	Type	Constraints
article_id	BIGINT	PK
title	VARCHAR(255)	UNIQUE, NOT NULL
slug	VARCHAR(255)	UNIQUE
content	TEXT	NOT NULL
category	VARCHAR(100)	"Sizing", "Returns", "Shipping", "Payment"
is_published	BOOLEAN	DEFAULT TRUE
view_count	INTEGER	DEFAULT 0
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

21. REVIEWS

Column	Type	Constraints
review_id	BIGINT	PK
product_id	UUID	FK -> products
order_id	UUID	FK -> orders (verify purchase)
customer_id	UUID	FK -> customers

Column	Type	Constraints
rating	INTEGER	NOT NULL, CHECK 1-5
title	VARCHAR(255)	
content	TEXT	
verified_purchase	BOOLEAN	DEFAULT TRUE
helpful_count	INTEGER	DEFAULT 0
images_json	JSONB	URLs to review photos
is_approved	BOOLEAN	DEFAULT FALSE
created_at	TIMESTAMP	NOT NULL

22. DAMAGE_ASSESSMENTS

Column	Type	Constraints
assessment_id	BIGINT	PK
return_id	UUID	FK -> returns
assessed_by_user_id	BIGINT	FK -> users (warehouse staff)
damage_level	ENUM('none','minor','moderate','severe','total_loss')	NOT NULL
damage_description	TEXT	
repair_feasible	BOOLEAN	
estimated_repair_cost_inr	DECIMAL(10,2)	
assessment_date	DATE	NOT NULL
photo_urls	TEXT[]	
created_at	TIMESTAMP	NOT NULL

23-27. Authorization & User Management

23. USERS (Internal staff)

Column	Type	Constraints
user_id	BIGINT	PK
email	VARCHAR(255)	UNIQUE, NOT NULL
password_hash	BYTEA	NOT NULL
full_name	VARCHAR(255)	NOT NULL
role_id	BIGINT	FK -> roles
department	ENUM('operations','support','warehouse','finance','admin')	
is_active	BOOLEAN	DEFAULT TRUE
last_login_at	TIMESTAMP	
created_at	TIMESTAMP	NOT NULL

24. ROLES

Column	Type	Constraints
role_id	BIGINT	PK
role_name	VARCHAR(50)	UNIQUE (admin, support_agent, warehouse_staff)
description	TEXT	

25. PERMISSIONS

Column	Type	Constraints
permission_id	BIGINT	PK
permission_name	VARCHAR(100)	UNIQUE (view_orders, approve_refund, access_pii)
description	TEXT	

26. USER_ROLES

Column	Type	Constraints
user_id	BIGINT	FK -> users
role_id	BIGINT	FK -> roles
PRIMARY KEY (user_id, role_id)		

27. ROLE_PERMISSIONS

Column	Type	Constraints
role_id	BIGINT	FK -> roles
permission_id	BIGINT	FK -> permissions
PRIMARY KEY (role_id, permission_id)		

28-32. Audit, Compliance & Configuration

28. AUDIT_LOG (Immutable)

Column	Type	Constraints
log_id	BIGINT	PK
agent_name	VARCHAR(100)	(catalog_agent, order_agent, etc.)
action_type	VARCHAR(100)	(search_products, create_order, approve_refund)
actor_type	ENUM('agent','user','system')	
actor_id	VARCHAR(255)	user_id or agent_id
customer_id	UUID	Nullable, for customer-facing actions
order_id	UUID	Nullable
entity_type	VARCHAR(100)	(product, order, refund)
entity_id	VARCHAR(255)	

Column	Type	Constraints
action_details	JSONB	Full context (old values, new values, decision logic)
result	ENUM('success','failure','manual_review_required')	
error_message	TEXT	If failure
tool_calls	JSONB	List of tools called, inputs, outputs
reasoning_notes	TEXT	Why agent made this decision
approval_status	ENUM('auto_approved','pending_approval','approved_by_human','rejected')	
approved_by_user_id	BIGINT	FK -> users (if manual approval)
created_at	TIMESTAMP	NOT NULL (millisecond precision)
created_at_ms	BIGINT	For sub-second ordering

29. COMPLIANCE_FLAGS

Column	Type	Constraints
flag_id	BIGINT	PK
flag_type	ENUM('unverified_claim','tax_violation','privacy_issue','policy_exception','legal_required')	
severity	ENUM('info','warning','critical')	
entity_type	VARCHAR(100)	(product, policy, email_content)
entity_id	VARCHAR(255)	
description	TEXT	
resolution_status	ENUM('open','reviewed','resolved','escalated')	
flagged_by_agent	VARCHAR(100)	
reviewed_by_user_id	BIGINT	FK -> users (nullable)
review_notes	TEXT	

Column	Type	Constraints
created_at	TIMESTAMP	NOT NULL
resolved_at	TIMESTAMP	

30. FRAUD_FLAGS

Column	Type	Constraints
fraud_flag_id	BIGINT	PK
order_id	UUID	FK -> orders (nullable)
customer_id	UUID	FK -> customers (nullable)
fraud_score	DECIMAL(3,2)	0.0-1.0
fraud_indicators	JSONB	Array of detected risk patterns
reason_codes	TEXT[]	(new_customer, high_value, unusual_address, velocity_check)
manual_review_status	ENUM('pending','reviewed','flagged_as_fraud','approved','false_positive')	
reviewed_by_user_id	BIGINT	FK -> users (nullable)
created_at	TIMESTAMP	NOT NULL

31. POLICIES (Versioned configuration)

Column	Type	Constraints
policy_id	BIGINT	PK
policy_type	ENUM('return','shipping','privacy','terms','payment','delivery')	NOT NULL
version	INTEGER	NOT NULL, version 1, 2, 3, etc.

Column	Type	Constraints
content	TEXT	Full policy text (HTML or Markdown)
effective_date	DATE	NOT NULL
expires_at	DATE	Nullable (NULL = current)
is_current	BOOLEAN	DEFAULT TRUE
created_by_user_id	BIGINT	FK -> users
approved_by_user_id	BIGINT	FK -> users
created_at	TIMESTAMP	NOT NULL

32. BUSINESS_CONFIG (Compliance, GST, shipping rules)

Column	Type	Constraints
config_id	BIGINT	PK
config_key	VARCHAR(255)	UNIQUE (free_shipping_threshold_inr, exchange_window_days)
config_value	VARCHAR(255) | JSONB	Value or JSON object
data_type	ENUM('integer','decimal','string','boolean','json')	
description	TEXT	
environment	ENUM('dev','staging','production')	
updated_by_user_id	BIGINT	FK -> users
requires_approval	BOOLEAN	(tax rates, refund policy = TRUE)
approval_status	ENUM('approved','pending_approval','rejected')	
approved_by_user_id	BIGINT	FK -> users (nullable)

Column	Type	Constraints
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Query Optimization: Indexes & Partitioning

Strategic Indexes by Access Pattern

Catalog Search (High-frequency, read-heavy)

```
CREATE INDEX idx_products_name_search ON products(name) WHERE is_active =
TRUE;
CREATE INDEX idx_products_category ON products(category_id) WHERE is_active =
TRUE;
CREATE INDEX idx_product_variants_sku ON product_variants(variant_sku);
CREATE INDEX idx_product_variants_color_size ON product_variants(color_name,
size);
CREATE INDEX idx_inventory_variant_quantity ON inventory(variant_id,
available_quantity) WHERE available_quantity > 0;
```

Customer Lookups

```
CREATE INDEX idx_customers_email ON customers(email) WHERE deleted_at IS
NULL;
CREATE INDEX idx_customers_phone ON customers(phone);
CREATE INDEX idx_addresses_customer ON addresses(customer_id);
CREATE INDEX idx_payment_methods_customer ON payment_methods(customer_id,
is_default);
```

Order & Transaction Queries

```
CREATE INDEX idx_orders_customer_date ON orders(customer_id, created_at
DESC);
CREATE INDEX idx_orders_status ON orders(order_status) INCLUDE (customer_id,
total_inr);
CREATE INDEX idx_order_items_order ON order_items(order_id);
CREATE INDEX idx_payments_order ON payments(order_id, payment_status);
CREATE INDEX idx_payments_gateway_ref ON payments(gateway_reference) WHERE
payment_status = 'confirmed';
CREATE INDEX idx_invoices_order ON invoices(order_id);
```

Fulfillment & Tracking

```
CREATE INDEX idx_shipments_order ON shipments(order_id);
CREATE INDEX idx_shipments_awb ON shipments(awb_number);
CREATE INDEX idx_shipments_status ON shipments(shipment_status) INCLUDE
(order_id, scheduled_delivery_date);
CREATE INDEX idx_shipment_tracking_shipment ON shipment_tracking(shipment_id,
event_timestamp DESC);
```

Returns & Refunds

```
CREATE INDEX idx_returns_order ON returns(order_id);
CREATE INDEX idx_returns_status ON returns(return_status) INCLUDE
(customer_id, created_at);
CREATE INDEX idx_return_items_return ON return_items(return_id);
CREATE INDEX idx_refunds_order ON refunds(order_id, refund_status);
```

Support & Communication

```
CREATE INDEX idx_support_tickets_customer ON support_tickets(customer_id,
created_at DESC);
CREATE INDEX idx_support_tickets_status ON support_tickets(ticket_status)
INCLUDE (priority, assigned_to_user_id);
CREATE INDEX idx_ticket_messages_ticket ON ticket_messages(ticket_id,
created_at);
```

Audit & Compliance

```
CREATE INDEX idx_audit_log_agent ON audit_log(agent_name, created_at_ms
DESC);
CREATE INDEX idx_audit_log_customer ON audit_log(customer_id, created_at
DESC);
CREATE INDEX idx_audit_log_entity ON audit_log(entity_type, entity_id,
created_at DESC);
CREATE INDEX idx_compliance_flags_type ON compliance_flags(flag_type,
severity);
CREATE INDEX idx_fraud_flags_customer ON fraud_flags(customer_id, created_at
DESC);
```

Full-Text Search (Via Elasticsearch integration, not PostgreSQL FTS)

```
-- Denormalized search table (optional, synced from PostgreSQL to
Elasticsearch)
CREATE INDEX idx_product_search_text ON products USING
GIN(to_tsvector('english', name || ' ' || description));
```

Partitioning Strategy

Time-Based Partitioning (Reduces scan time for historical queries)

Orders Table (by month, for 2+ years rolling)

```
CREATE TABLE orders (
  ...
) PARTITION BY RANGE (DATE_TRUNC('month', created_at));

CREATE TABLE orders_2026_10 PARTITION OF orders
  FOR VALUES FROM ('2026-10-01') TO ('2026-11-01');
CREATE TABLE orders_2026_11 PARTITION OF orders
  FOR VALUES FROM ('2026-11-01') TO ('2026-12-01');
```

Audit Log (critical for performance; growing rapidly)

```
CREATE TABLE audit_log (
  ...
) PARTITION BY RANGE (DATE_TRUNC('week', created_at));
```

Shipment Tracking (queries recent events often, old events rarely)

```
CREATE TABLE shipment_tracking (
  ...
) PARTITION BY RANGE (DATE_TRUNC('month', event_timestamp));
```

Query Performance Guidelines

Query Type	Expected Time	Index Strategy
Product search (by name/category)	<50ms	Covering indexes (include rating, price)
Customer orders (last 10)	<20ms	(customer_id, created_at DESC)

Query Type	Expected Time	Index Strategy
Payment lookup (by gateway ref)	<10ms	Unique index on gateway_reference
Shipment tracking (live)	<30ms	shipment_id + event_timestamp
Audit log query (7-day)	<100ms	agent_name + partition pruning
Compliance flag search	<200ms	flag_type + severity

Vacuum & Maintenance

```
-- Weekly VACUUM ANALYZE for transactional tables
VACUUM ANALYZE orders, order_items, payments, shipments, returns;

-- Monthly REINDEX for frequently-updated tables
REINDEX INDEX CONCURRENTLY idx_orders_customer_date;
REINDEX INDEX CONCURRENTLY idx_audit_log_agent;
```

Caching at Redis Layer (Reduces database load)

- Product details: Cache key `product:{product_id}`, TTL 24 hours
- Product variants: Cache key `variant:{variant_id}`, TTL 24 hours
- Customer session: Cache key `session:{session_id}`, TTL 30 minutes
- Cart state: Cache key `cart:{customer_id}`, TTL 24 hours
- Inventory levels: Cache key `inventory:{variant_id}`, TTL 5 minutes (frequently changing)
- Business config: Cache key `config:{key}`, TTL 6 hours

Complete DDL (Data Definition Language)

Download and run this SQL to initialize the schema.

```
--
================================================================================
-- NOIR ATELIER DATABASE SCHEMA (PostgreSQL 16+)
-- Production-Ready, Audit-Complete, Compliance-First
--
================================================================================
```

```
-- Enable extensions
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "plpgsql";

--
================================================================================
-- 1. CORE ENTITIES
--
================================================================================

CREATE TABLE customers (
  customer_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email VARCHAR(255) UNIQUE NOT NULL,
  phone VARCHAR(15) UNIQUE,
  password_hash BYTEA NOT NULL,
  first_name VARCHAR(100) NOT NULL,
  last_name VARCHAR(100),
  date_of_birth DATE,
  gender VARCHAR(20),
  email_verified BOOLEAN DEFAULT FALSE,
  phone_verified BOOLEAN DEFAULT FALSE,
  newsletter_opt_in BOOLEAN DEFAULT FALSE,
  marketing_consent BOOLEAN DEFAULT FALSE,
  preferred_language VARCHAR(5) DEFAULT 'en',
  loyalty_points INTEGER DEFAULT 0,
  vip_status VARCHAR(20) DEFAULT 'standard',
  last_login_at TIMESTAMP,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  deleted_at TIMESTAMP
);

CREATE INDEX idx_customers_email ON customers(email) WHERE deleted_at IS
NULL;
CREATE INDEX idx_customers_phone ON customers(phone);

CREATE TABLE addresses (
  address_id BIGSERIAL PRIMARY KEY,
  customer_id UUID NOT NULL REFERENCES customers(customer_id),
  address_type VARCHAR(20) NOT NULL,
  full_name VARCHAR(100) NOT NULL,
```

```
  street_address VARCHAR(255) NOT NULL,
  apartment_floor VARCHAR(100),
  locality VARCHAR(100) NOT NULL,
  city VARCHAR(100) NOT NULL,
  state VARCHAR(100) NOT NULL,
  postal_code VARCHAR(10) NOT NULL,
  country VARCHAR(2) DEFAULT 'IN',
  phone VARCHAR(15) NOT NULL,
  landmark VARCHAR(100),
  google_maps_url TEXT,
  latitude DECIMAL(10,8),
  longitude DECIMAL(11,8),
  is_default_shipping BOOLEAN DEFAULT FALSE,
  is_serviceable BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_addresses_customer ON addresses(customer_id);

CREATE TABLE payment_methods (
  payment_method_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  customer_id UUID NOT NULL REFERENCES customers(customer_id),
  method_type VARCHAR(20) NOT NULL,
  nickname VARCHAR(50),
  is_default BOOLEAN DEFAULT FALSE,
  token VARCHAR(255) NOT NULL,
  last_four VARCHAR(4),
  card_network VARCHAR(20),
  expiry_date VARCHAR(5),
  bank_name VARCHAR(100),
  account_number BYTEA,
  ifsc_code VARCHAR(11),
  verified BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  deleted_at TIMESTAMP
);

CREATE INDEX idx_payment_methods_customer ON payment_methods(customer_id);

CREATE TABLE categories (
  category_id BIGSERIAL PRIMARY KEY,
  name VARCHAR(100) UNIQUE NOT NULL,
```

```
  slug VARCHAR(100) UNIQUE,
  description TEXT,
  parent_category_id BIGINT REFERENCES categories(category_id),
  image_url TEXT,
  display_order INTEGER,
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE products (
  product_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  sku VARCHAR(50) UNIQUE NOT NULL,
  name VARCHAR(255) NOT NULL,
  slug VARCHAR(255) UNIQUE,
  description TEXT,
  short_description VARCHAR(500),
  category_id BIGINT NOT NULL REFERENCES categories(category_id),
  subcategory_id BIGINT REFERENCES categories(category_id),
  price_inr DECIMAL(10,2) NOT NULL CHECK (price_inr > 0),
  compare_at_price_inr DECIMAL(10,2),
  cost_price_inr DECIMAL(10,2),
  gst_rate DECIMAL(4,2) DEFAULT 5.0,
  gst_hsn_code VARCHAR(8),
  material VARCHAR(100),
  material_percentage JSONB,
  care_instructions TEXT,
  brand VARCHAR(100),
  designer_name VARCHAR(100),
  made_in_country VARCHAR(2),
  made_in_city VARCHAR(100),
  manufacturing_date DATE,
  weight_grams INTEGER,
  dimensions_length_cm DECIMAL(6,2),
  dimensions_width_cm DECIMAL(6,2),
  dimensions_height_cm DECIMAL(6,2),
  is_active BOOLEAN DEFAULT TRUE,
  is_featured BOOLEAN DEFAULT FALSE,
  bestseller_rank INTEGER,
  rating DECIMAL(3,2) DEFAULT 0,
  review_count INTEGER DEFAULT 0,
  external_product_url TEXT,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
```

```
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  deleted_at TIMESTAMP
);

CREATE INDEX idx_products_name_search ON products(name) WHERE is_active =
TRUE;
CREATE INDEX idx_products_category ON products(category_id) WHERE is_active =
TRUE;

CREATE TABLE product_variants (
  variant_id BIGSERIAL PRIMARY KEY,
  product_id UUID NOT NULL REFERENCES products(product_id),
  variant_sku VARCHAR(50) UNIQUE NOT NULL,
  color_name VARCHAR(50) NOT NULL,
  color_hex VARCHAR(7),
  size VARCHAR(10) NOT NULL,
  size_chart_url TEXT,
  price_inr DECIMAL(10,2),
  quantity_on_hand INTEGER DEFAULT 0,
  variant_image_url TEXT,
  is_available BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_product_variants_sku ON product_variants(variant_sku);
CREATE INDEX idx_product_variants_product ON product_variants(product_id);

CREATE TABLE inventory (
  inventory_id BIGSERIAL PRIMARY KEY,
  variant_id BIGINT UNIQUE NOT NULL REFERENCES product_variants(variant_id),
  quantity_on_hand INTEGER NOT NULL DEFAULT 0,
  reserved_quantity INTEGER DEFAULT 0,
  reorder_level INTEGER DEFAULT 10,
  reorder_quantity INTEGER DEFAULT 50,
  supplier_id BIGINT,
  lead_time_days INTEGER,
  last_reordered_at TIMESTAMP,
  next_reorder_date DATE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE INDEX idx_inventory_variant_quantity ON inventory(variant_id,
quantity_on_hand);

--
================================================================================
-- 2. ORDERS & TRANSACTIONS
--
================================================================================

CREATE TABLE orders (
  order_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  order_number VARCHAR(20) UNIQUE NOT NULL,
  customer_id UUID NOT NULL REFERENCES customers(customer_id),
  order_status VARCHAR(50) DEFAULT 'pending_payment' NOT NULL,
  subtotal_inr DECIMAL(10,2) NOT NULL,
  tax_inr DECIMAL(10,2) DEFAULT 0,
  shipping_charge_inr DECIMAL(10,2) DEFAULT 0,
  discount_amount_inr DECIMAL(10,2) DEFAULT 0,
  total_inr DECIMAL(10,2) NOT NULL,
  currency VARCHAR(3) DEFAULT 'INR',
  shipping_address_id BIGINT NOT NULL REFERENCES addresses(address_id),
  billing_address_id BIGINT REFERENCES addresses(address_id),
  delivery_date_estimate DATE,
  actual_delivery_date DATE,
  notes TEXT,
  agent_notes TEXT,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  confirmed_at TIMESTAMP,
  shipped_at TIMESTAMP,
  delivered_at TIMESTAMP,
  cancelled_at TIMESTAMP,
  deleted_at TIMESTAMP
);

CREATE INDEX idx_orders_customer_date ON orders(customer_id, created_at
DESC);
CREATE INDEX idx_orders_status ON orders(order_status);

CREATE TABLE order_items (
  order_item_id BIGSERIAL PRIMARY KEY,
  order_id UUID NOT NULL REFERENCES orders(order_id),
  variant_id BIGINT NOT NULL REFERENCES product_variants(variant_id),
```

```
  product_name VARCHAR(255) NOT NULL,
  sku VARCHAR(50) NOT NULL,
  color VARCHAR(50),
  size VARCHAR(10),
  quantity INTEGER NOT NULL CHECK (quantity > 0),
  unit_price_inr DECIMAL(10,2) NOT NULL,
  tax_inr DECIMAL(10,2) DEFAULT 0,
  discount_inr DECIMAL(10,2) DEFAULT 0,
  total_inr DECIMAL(10,2) NOT NULL,
  return_status VARCHAR(50) DEFAULT 'none',
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_order_items_order ON order_items(order_id);

CREATE TABLE payments (
  payment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  order_id UUID NOT NULL REFERENCES orders(order_id),
  payment_method_type VARCHAR(50) NOT NULL,
  payment_method_id UUID REFERENCES payment_methods(payment_method_id),
  amount_inr DECIMAL(10,2) NOT NULL CHECK (amount_inr > 0),
  payment_status VARCHAR(50) DEFAULT 'pending' NOT NULL,
  gateway_reference VARCHAR(255),
  gateway_response JSONB,
  bank_transaction_id VARCHAR(255),
  upi_transaction_id VARCHAR(255),
  verification_status VARCHAR(50) DEFAULT 'pending_verification',
  verified_by VARCHAR(255),
  verified_at TIMESTAMP,
  error_message TEXT,
  retry_count INTEGER DEFAULT 0,
  next_retry_at TIMESTAMP,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  completed_at TIMESTAMP
);

CREATE INDEX idx_payments_order ON payments(order_id);
CREATE INDEX idx_payments_gateway_ref ON payments(gateway_reference);

CREATE TABLE invoices (
  invoice_id BIGSERIAL PRIMARY KEY,
  order_id UUID UNIQUE NOT NULL REFERENCES orders(order_id),
  invoice_number VARCHAR(20) UNIQUE,
```

```
  invoice_date DATE NOT NULL,
  gst_invoice_type VARCHAR(20),
  seller_name VARCHAR(255) NOT NULL,
  seller_gstin VARCHAR(15),
  seller_address TEXT NOT NULL,
  billing_customer_name VARCHAR(255) NOT NULL,
  billing_customer_gstin VARCHAR(15),
  billing_customer_address TEXT NOT NULL,
  shipping_address TEXT,
  line_items_json JSONB,
  subtotal_inr DECIMAL(10,2) NOT NULL,
  sgst_inr DECIMAL(10,2) DEFAULT 0,
  cgst_inr DECIMAL(10,2) DEFAULT 0,
  igst_inr DECIMAL(10,2) DEFAULT 0,
  total_tax_inr DECIMAL(10,2) NOT NULL,
  total_inr DECIMAL(10,2) NOT NULL,
  qr_code TEXT,
  irn VARCHAR(64),
  ack_num VARCHAR(64),
  ack_date_time TIMESTAMP,
  is_generated BOOLEAN DEFAULT FALSE,
  generated_at TIMESTAMP,
  pdf_url TEXT,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_invoices_order ON invoices(order_id);

--
================================================================================
-- 3. FULFILLMENT & SHIPPING
--
================================================================================

CREATE TABLE courier_partners (
  courier_id BIGSERIAL PRIMARY KEY,
  name VARCHAR(100) UNIQUE NOT NULL,
  api_key VARCHAR(255),
  api_secret VARCHAR(255),
  api_base_url TEXT,
  is_active BOOLEAN DEFAULT TRUE,
  priority_rank INTEGER,
  serviceable_pincodes_json JSONB,
```

```
  rate_per_kg_inr DECIMAL(10,2),
  cod_fee_percent DECIMAL(5,2),
  average_delivery_days INTEGER,
  last_sync_at TIMESTAMP,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE shipments (
  shipment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  order_id UUID NOT NULL REFERENCES orders(order_id),
  courier_partner_id BIGINT NOT NULL REFERENCES courier_partners(courier_id),
  awb_number VARCHAR(50) UNIQUE,
  tracking_url TEXT,
  shipment_status VARCHAR(50) DEFAULT 'created',
  weight_grams INTEGER,
  dimensions_json JSONB,
  shipment_charge_inr DECIMAL(10,2),
  shipping_address_id BIGINT NOT NULL REFERENCES addresses(address_id),
  scheduled_pickup_date DATE,
  actual_pickup_date DATE,
  scheduled_delivery_date DATE,
  actual_delivery_date DATE,
  signature_proof_url TEXT,
  delivery_notes TEXT,
  failed_delivery_reason TEXT,
  return_to_warehouse BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_shipments_order ON shipments(order_id);
CREATE INDEX idx_shipments_awb ON shipments(awb_number);
CREATE INDEX idx_shipments_status ON shipments(shipment_status);

CREATE TABLE shipment_tracking (
  tracking_id BIGSERIAL PRIMARY KEY,
  shipment_id UUID NOT NULL REFERENCES shipments(shipment_id),
  event_type VARCHAR(50) NOT NULL,
  event_description TEXT NOT NULL,
  event_location VARCHAR(255),
  latitude DECIMAL(10,8),
  longitude DECIMAL(11,8),
  event_timestamp TIMESTAMP NOT NULL,
```

```
  received_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  courier_event_id VARCHAR(255),
  status_image_url TEXT
);

CREATE INDEX idx_shipment_tracking_shipment ON shipment_tracking(shipment_id,
event_timestamp DESC);

--
================================================================================
-- 4. RETURNS & REFUNDS
--
================================================================================

CREATE TABLE returns (
  return_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  order_id UUID NOT NULL REFERENCES orders(order_id),
  return_request_type VARCHAR(50) NOT NULL,
  return_reason VARCHAR(50) NOT NULL,
  return_reason_details TEXT,
  return_status VARCHAR(50) DEFAULT 'requested',
  eligibility_check VARCHAR(50),
  days_since_delivery INTEGER,
  pickup_date DATE,
  return_received_date DATE,
  inspection_date DATE,
  inspection_result VARCHAR(50),
  approved_remedy VARCHAR(50),
  replacement_variant_id BIGINT REFERENCES product_variants(variant_id),
  replacement_order_id UUID REFERENCES orders(order_id),
  refund_approved_amount_inr DECIMAL(10,2),
  refund_reason_code VARCHAR(50),
  deduction_amount_inr DECIMAL(10,2) DEFAULT 0,
  notes TEXT,
  photo_urls TEXT[],
  return_request_date DATE NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  completed_at TIMESTAMP
);

CREATE INDEX idx_returns_order ON returns(order_id);
CREATE INDEX idx_returns_status ON returns(return_status);
```

```
CREATE TABLE return_items (
  return_item_id BIGSERIAL PRIMARY KEY,
  return_id UUID NOT NULL REFERENCES returns(return_id),
  order_item_id BIGINT NOT NULL REFERENCES order_items(order_item_id),
  quantity_returned INTEGER NOT NULL,
  unit_refund_amount_inr DECIMAL(10,2),
  total_refund_amount_inr DECIMAL(10,2),
  item_condition VARCHAR(50),
  item_notes TEXT
);

CREATE TABLE refunds (
  refund_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  return_id UUID REFERENCES returns(return_id),
  order_id UUID NOT NULL REFERENCES orders(order_id),
  refund_amount_inr DECIMAL(10,2) NOT NULL,
  refund_reason VARCHAR(50) NOT NULL,
  refund_status VARCHAR(50) DEFAULT 'initiated',
  original_payment_method_id UUID REFERENCES
payment_methods(payment_method_id),
  refund_gateway_reference VARCHAR(255),
  refund_bank_transaction_id VARCHAR(255),
  refund_initiated_at TIMESTAMP NOT NULL,
  refund_completed_at TIMESTAMP,
  processing_fee_inr DECIMAL(10,2) DEFAULT 0,
  notes TEXT,
  approved_by VARCHAR(255),
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_refunds_order ON refunds(order_id);

--
============================================================================
-- 5. SUPPORT & COMMUNICATION
--
============================================================================

CREATE TABLE support_tickets (
  ticket_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  customer_id UUID NOT NULL REFERENCES customers(customer_id),
  order_id UUID REFERENCES orders(order_id),
  subject VARCHAR(255) NOT NULL,
```

```
  description TEXT NOT NULL,
  ticket_status VARCHAR(50) DEFAULT 'open',
  priority VARCHAR(50) DEFAULT 'medium',
  category VARCHAR(50) NOT NULL,
  assigned_to_user_id BIGINT,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  resolved_at TIMESTAMP,
  resolution_notes TEXT
);

CREATE INDEX idx_support_tickets_customer ON support_tickets(customer_id);
CREATE INDEX idx_support_tickets_status ON support_tickets(ticket_status);

CREATE TABLE ticket_messages (
  message_id BIGSERIAL PRIMARY KEY,
  ticket_id UUID NOT NULL REFERENCES support_tickets(ticket_id),
  sender_type VARCHAR(20) NOT NULL,
  sender_user_id UUID,
  message_text TEXT NOT NULL,
  attachments_json JSONB,
  is_public BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_ticket_messages_ticket ON ticket_messages(ticket_id);

CREATE TABLE reviews (
  review_id BIGSERIAL PRIMARY KEY,
  product_id UUID NOT NULL REFERENCES products(product_id),
  order_id UUID NOT NULL REFERENCES orders(order_id),
  customer_id UUID NOT NULL REFERENCES customers(customer_id),
  rating INTEGER NOT NULL CHECK (rating BETWEEN 1 AND 5),
  title VARCHAR(255),
  content TEXT,
  verified_purchase BOOLEAN DEFAULT TRUE,
  helpful_count INTEGER DEFAULT 0,
  images_json JSONB,
  is_approved BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_reviews_product ON reviews(product_id);
```

```
--
================================================================================
-- 6. AUTHORIZATION & AUDIT
--
================================================================================

CREATE TABLE users (
  user_id BIGSERIAL PRIMARY KEY,
  email VARCHAR(255) UNIQUE NOT NULL,
  password_hash BYTEA NOT NULL,
  full_name VARCHAR(255) NOT NULL,
  department VARCHAR(50),
  is_active BOOLEAN DEFAULT TRUE,
  last_login_at TIMESTAMP,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE roles (
  role_id BIGSERIAL PRIMARY KEY,
  role_name VARCHAR(50) UNIQUE,
  description TEXT
);

CREATE TABLE permissions (
  permission_id BIGSERIAL PRIMARY KEY,
  permission_name VARCHAR(100) UNIQUE,
  description TEXT
);

CREATE TABLE user_roles (
  user_id BIGINT REFERENCES users(user_id) ON DELETE CASCADE,
  role_id BIGINT REFERENCES roles(role_id) ON DELETE CASCADE,
  PRIMARY KEY (user_id, role_id)
);

CREATE TABLE role_permissions (
  role_id BIGINT REFERENCES roles(role_id) ON DELETE CASCADE,
  permission_id BIGINT REFERENCES permissions(permission_id) ON DELETE
CASCADE,
  PRIMARY KEY (role_id, permission_id)
);
```

```
CREATE TABLE audit_log (
  log_id BIGSERIAL PRIMARY KEY,
  agent_name VARCHAR(100),
  action_type VARCHAR(100),
  actor_type VARCHAR(50),
  actor_id VARCHAR(255),
  customer_id UUID,
  order_id UUID,
  entity_type VARCHAR(100),
  entity_id VARCHAR(255),
  action_details JSONB,
  result VARCHAR(50),
  error_message TEXT,
  tool_calls JSONB,
  reasoning_notes TEXT,
  approval_status VARCHAR(50),
  approved_by_user_id BIGINT REFERENCES users(user_id),
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  created_at_ms BIGINT DEFAULT EXTRACT(EPOCH FROM CURRENT_TIMESTAMP) * 1000
);

CREATE INDEX idx_audit_log_agent ON audit_log(agent_name, created_at DESC);
CREATE INDEX idx_audit_log_customer ON audit_log(customer_id, created_at
DESC);

CREATE TABLE compliance_flags (
  flag_id BIGSERIAL PRIMARY KEY,
  flag_type VARCHAR(50),
  severity VARCHAR(50),
  entity_type VARCHAR(100),
  entity_id VARCHAR(255),
  description TEXT,
  resolution_status VARCHAR(50),
  flagged_by_agent VARCHAR(100),
  reviewed_by_user_id BIGINT REFERENCES users(user_id),
  review_notes TEXT,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  resolved_at TIMESTAMP
);

CREATE TABLE fraud_flags (
  fraud_flag_id BIGSERIAL PRIMARY KEY,
  order_id UUID REFERENCES orders(order_id),
```

```
  customer_id UUID REFERENCES customers(customer_id),
  fraud_score DECIMAL(3,2),
  fraud_indicators JSONB,
  reason_codes TEXT[],
  manual_review_status VARCHAR(50),
  reviewed_by_user_id BIGINT REFERENCES users(user_id),
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

--
================================================================================
-- 7. CONFIGURATION
--
================================================================================

CREATE TABLE policies (
  policy_id BIGSERIAL PRIMARY KEY,
  policy_type VARCHAR(50) NOT NULL,
  version INTEGER NOT NULL,
  content TEXT,
  effective_date DATE NOT NULL,
  expires_at DATE,
  is_current BOOLEAN DEFAULT TRUE,
  created_by_user_id BIGINT REFERENCES users(user_id),
  approved_by_user_id BIGINT REFERENCES users(user_id),
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE business_config (
  config_id BIGSERIAL PRIMARY KEY,
  config_key VARCHAR(255) UNIQUE,
  config_value TEXT,
  data_type VARCHAR(50),
  description TEXT,
  environment VARCHAR(50),
  updated_by_user_id BIGINT REFERENCES users(user_id),
  requires_approval BOOLEAN,
  approval_status VARCHAR(50),
  approved_by_user_id BIGINT REFERENCES users(user_id),
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE TABLE knowledge_base (
  article_id BIGSERIAL PRIMARY KEY,
  title VARCHAR(255) UNIQUE NOT NULL,
  slug VARCHAR(255) UNIQUE,
  content TEXT NOT NULL,
  category VARCHAR(100),
  is_published BOOLEAN DEFAULT TRUE,
  view_count INTEGER DEFAULT 0,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE damage_assessments (
  assessment_id BIGSERIAL PRIMARY KEY,
  return_id UUID NOT NULL REFERENCES returns(return_id),
  assessed_by_user_id BIGINT REFERENCES users(user_id),
  damage_level VARCHAR(50),
  damage_description TEXT,
  repair_feasible BOOLEAN,
  estimated_repair_cost_inr DECIMAL(10,2),
  assessment_date DATE NOT NULL,
  photo_urls TEXT[],
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

--
================================================================================
-- GRANT PERMISSIONS (Adjust to your user/role model)
--
================================================================================

-- Example: create app user with limited permissions
CREATE ROLE noir_app WITH LOGIN PASSWORD 'secure_password';
GRANT CONNECT ON DATABASE noir_atelier TO noir_app;
GRANT USAGE ON SCHEMA public TO noir_app;
GRANT SELECT, INSERT, UPDATE ON ALL TABLES IN SCHEMA public TO noir_app;
GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public TO noir_app;
```

How to Use

1. Create the database: `createdb noir_atelier`
2. Run the DDL above: `psql -U postgres -d noir_atelier -f schema.sql`

3. Verify tables: `\dt` in psql
4. Run indexes: Included in DDL
5. Configure app connection string:
   `postgresql://noir_app:password@localhost:5432/noir_atelier`